15.8 (1). $S = E \times H \times \frac{1}{2}$

$= \frac{1}{2} E \cdot \frac{\sqrt{\varepsilon_0}}{\sqrt{\mu_0}} E = 1.4 \times 10^3 W/m^3$

$E = \sqrt{\frac{2S}{\sqrt{\varepsilon_0}} \cdot \sqrt{\mu_0}} = 1.03 \times 10^3 \times 10^3 V/m.$

$B = \mu_0 H = \mu_0 \frac{\sqrt{\varepsilon_0}}{\sqrt{\mu_0}} \cdot E$

$= \sqrt{\varepsilon_0 \mu_0} \cdot E = 1.03 \times 10^3 \times \sqrt{4\pi \times 10^{-7} \times 8.85 \times 10^{-12}}$

$= 3.4 \times 10^{-6} T.$

(2) $P = S \cdot 4\pi r^2$

$= 1.4 \times 10^3 W/m^2 \times 4\pi \times (1.5 \times 10^{11} m)^2$

$= 3.96 \times 10^{26} W.$

15.11 (1) $f = \frac{c}{\lambda} = \frac{3 \times 10^8 m/s}{3.0 m} = 10^8 Hz.$

(2). B的方向沿z轴正方向.

$B = \mu H$

$= \sqrt{\mu_0 \varepsilon_0} E = \sqrt{4\pi \times 10^{-7} \times 8.85 \times 10^{-12}} \times 300$

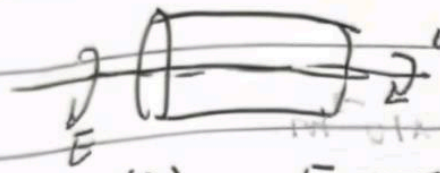
$= 1 \times 10^{-6} T.$

(3). $\omega = 2\pi f = 2\pi \times 10^8 Hz = 6.28 \times 10^8 rad/s.$

(4). $S = \frac{1}{2} \sqrt{\frac{\varepsilon_0}{\mu_0}} \cdot E^2 = 119.4 W.$

15.14 (1). $I = 2\pi RL\cdot\sigma\frac{\omega}{2\pi} = RL\sigma\omega = RL\sigma\alpha t$

$$B = \mu_0 n I = \mu_0\cdot\frac{RL\alpha t\cdot\sigma}{L} = \mu_0 R\alpha t\cdot\sigma$$



(2). $$E\cdot 2\pi R = -\frac{d\Phi}{dt} = -\frac{d(\mu_0 R\alpha t\cdot\pi R^2\sigma)}{dt} = -\mu_0 R\alpha\cdot\pi R^2\sigma$$

$$E = \frac{-\mu_0 R\alpha\cdot\pi R^2\sigma}{2\pi R} = -\frac{\mu_0\alpha R^2\sigma}{2}$$

涡旋电场产生磁场方向阻碍原磁场变化.

(3). $$S = E\times H = \frac{B}{\mu_0}\cdot E = \frac{\mu_0 R\alpha t\sigma\cdot\frac{\mu_0\alpha R^2\sigma}{2}}{\mu_0} = \frac{\mu_0 R^3\alpha^2\sigma^2 t}{2}$$

(4). $$S\times 2\pi R\cdot L = \pi\mu_0 R^4\alpha^2\sigma^2 t L$$

$$\frac{d}{dt}\left(\frac{\pi R^2 L}{2\mu_0}B^2\right) = \frac{d}{dt}\left(\frac{\pi R^2 L}{2\mu_0}\cdot\mu_0^2 R^2\alpha^2 t^2\sigma^2\right)$$

$$= \frac{\pi R^2 L}{2\mu_0}\mu_0^2\cdot R^2\alpha^2\sigma^2\cdot 2t$$

$$= \pi\mu_0 R^4\alpha^2\sigma^2 t L.$$

则 S 圆筒面积分 $\frac{d}{dt}\left(\frac{\pi R^2 L}{2\mu_0}B^2\right)$, 得证.

19.2 解：

$$\begin{cases} \frac{1}{s_1}+\frac{n}{s_1'}=\frac{n-1}{R_1} \\ \frac{n}{s_2}+\frac{1}{s_2'}=\frac{1-n}{R_2} \\ s_2=-s_1' \end{cases}$$

解得 $\frac{1}{s_1}+\frac{1}{s_2'}=0.52\times\frac{2}{0.1}=10.4$.

$$f=\frac{1}{\frac{1}{s_1}+\frac{1}{s_2'}}=\frac{1}{10.4}=9.6\times10^{-2}m.$$

发散：

$$\text{则}f=-\frac{1}{\frac{1}{s_1}+\frac{1}{s_2'}}=-\frac{1}{10.4}=-9.6\times10^{-2}m.$$

19.3 (1) $\frac{1}{f}=(n-1)\times\left(\frac{1}{R_1}-\frac{1}{R_2}\right)=(n-1)\times\frac{1}{R_1}$

$$\text{则}f=\frac{R_1}{n-1}=\frac{57.1\times10^{-3}m}{0.2}=0.286m.$$

(2). $\frac{1}{s'}-\frac{1}{s}=\frac{1}{f}$.

$$s=0.5m.\ s'=\frac{1}{\frac{1}{0.286m}-\frac{1}{0.5m}}=0.668m$$

19.6. $s'=23cm-2cm=21cm.$

$$\frac{1}{s}+\frac{1}{s'}=\frac{1}{f_1}=\frac{1}{1cm}.$$

$$\text{则}s=1.05cm.$$

$$M=\frac{-25cm}{2cm}\times\frac{21cm}{1cm}=-262.5$$